

Understanding Analysis, 2nd Edition

Chapter 7 - Section 7.2 - The Riemann Integral

1. Let f be a bounded function on $[a, b]$, and let P be an arbitrary partition of $[a, b]$. First, explain why $U(f) \geq L(f, P)$. Now, prove Lemma 7.2.6: for any bounded function f on $[a, b]$, it is always the case that $U(f) \geq L(f)$.

Solution: We'll prove it by contradiction. Suppose that $U(f) < L(f, P)$. By definition of infimum, there must exist a partition P_2 such that $U(f, P_2) < L(f, P)$, for otherwise the infimum would've been at least $L(f, P)$.

But $U(f, P_2) < L(f, P)$ contradicts Lemma 7.2.4: $L(f, P_1) \leq U(f, P_2)$ for any partitions P_1, P_2 .

To use this fact to prove Lemma 7.2.6, note that $L(f)$ is the supremum of all lower sums while $U(f)$ is an upper bound. Since by definition, an upper bound is *at least* the supremum, it follows that $U(f) \geq L(f)$.

Alternatively, we can prove Lemma 7.2.6 directly by contradiction as well. Suppose $U(f) < L(f)$ and let m be the midpoint i.e. $m = \frac{L(f) + U(f)}{2}$. By definition of infimum, there exists a partition P_1 such that $U(f, P_1) < m$. By definition of supremum, there exists a partition P_2 such that $m < L(f, P_2)$. But $U(f, P_1) < m < L(f, P_2)$ contradicts Lemma 7.2.4.

2. Consider $f(x) = 1/x$ over the interval $[1, 4]$. Let P be the partition consisting of the points $\{1, 3/2, 2, 4\}$.
 - (a) Compute $L(f, P)$, $U(f, P)$ and $U(f, P) - L(f, P)$.
 - (b) What happens to the value of $U(f, P) - L(f, P)$ when we add the point 3 to the partition?
 - (c) Find a partition P' of $[1, 4]$ for which $U(f, P') - L(f, P') < 2/5$.

Solution:

$$(a) \quad L(f, P) = (2/3 * 1/2) + (1/2 * 1/2) + (1/4 * 2) = 13/12.$$

$$U(f, P) = (1 * 1/2) + (2/3 * 1/2) + (1/2 * 2) = 11/6.$$

$$U(f, P) - L(f, P) = 3/4.$$

- (b) It gets smaller, because adding a new point to a partition decreases the upper sum while increasing the lower sum (Lemma 7.2.3). The new values are:

$$L(f, P) = (2/3 * 1/2) + (1/2 * 1/2) + (1/3 * 1) + (1/4 * 1) = 14/12.$$

$$U(f, P) = (1 * 1/2) + (2/3 * 1/2) + (1/2 * 1) + (1/3 * 1) = 5/3.$$

$$U(f, P) - L(f, P) = 1/2.$$

- (c) I initially solved this by adding more points to the initial partition + some random guessing, but chatgpt suggested a smarter way.

Let's create an equally-spaced partition. Then we have

$$U - L = \Delta x \sum_{i=1}^n M_i - m_i$$

For $f(x) = 1/x$, the supremum $M_i = f(x_{i-1})$ and the infimum $m_i = f(x_i)$.

The sum telescopes:

$$\begin{aligned} U - L &= \Delta x [f(x_0) - f(x_1) + f(x_1) - f(x_2) + \dots + f(x_{n-1}) - f(x_n)] \\ &= \Delta x (f(x_0) - f(x_n)) \end{aligned}$$

We have $x_0 = 1$ and $x_n = 4$, resulting in $\Delta x(1 - 4) = \frac{3}{4}\Delta x$ and we want this to be less than $\frac{2}{5}$:

$$\frac{3}{4}\Delta x < \frac{2}{5} \implies \Delta x < \frac{8}{15}$$

We can pick $\Delta x = \frac{1}{2}$. So the final partition P' is $\{1, \frac{3}{2}, 2, \frac{5}{2}, 3, \frac{7}{2}, 4\}$.

3. (Sequential Criterion for Integrability)

- (a) Prove that a bounded function f is integrable on $[a, b]$ if and only if there exists a sequence of partitions $(P_n)_{n=1}^{\infty}$ satisfying

$$\lim_{n \rightarrow \infty} [U(f, P_n) - L(f, P_n)] = 0,$$

and in this case $\int_a^b f = \lim_{n \rightarrow \infty} U(f, P_n) = \lim_{n \rightarrow \infty} L(f, P_n)$.

- (b) For each n , let P_n be the partition of $[0, 1]$ into n equal subintervals. Find formulas for $U(f, P_n)$ and $L(f, P_n)$ if $f(x) = x$. The formula $1 + 2 + 3 + \dots + n = n(n+1)/2$ will be helpful.
- (c) Use the sequential criterion for integrability from (a) to show directly that $f(x) = x$ is integrable on $[0, 1]$ and compute $\int_0^1 f$.

Solution:

- (a) We'll prove the equivalence with the criterion of Theorem 7.2.8: for any $\epsilon > 0$ there exists a partition P_ϵ of $[a, b]$ such that

$$U(f, P_\epsilon) - L(f, P_\epsilon) < \epsilon$$

For the sufficiency direction, assume the sequential criterion is true. Then by definition of sequential convergence, for any $\epsilon > 0$, there exists N such that for all $n \geq N$

$$U(f, P_n) - L(f, P_n) < \epsilon$$

Then just set $P_\epsilon = P_N$.

For the necessity direction, assume Theorem 7.2.8 is true. Then we can build a sequence of partitions (P_n) where

$$U(f, P_n) - L(f, P_n) < 1/n$$

And since $\lim_{n \rightarrow \infty} 1/n = 0$, then $\lim_{n \rightarrow \infty} U(f, P_n) - L(f, P_n) = 0$ as well.

— **This is incorrect** —

Finally, to prove that $\int_a^b f = \lim_{n \rightarrow \infty} U(f, P_n) = \lim_{n \rightarrow \infty} L(f, P_n)$, we need to prove that both limits are equal to $U(f)$ and $L(f)$ according to definition 7.2.1 of the Riemann Integral.

Note that by exercise 1 and the fact that $U(f)$ and $L(f)$ are infimum / supremum:

$$\lim_{n \rightarrow \infty} U(f, P_n) \geq U(f) \geq L(f) \geq \lim_{n \rightarrow \infty} L(f, P_n)$$

Since we know $\lim_{n \rightarrow \infty} U(f, P_n) = \lim_{n \rightarrow \infty} L(f, P_n)$, the only solution to the inequality is if they're equal $U(f)$ and $L(f)$.

— **Mistake pointed out by chatgpt. Correction by my own.** —

The mistake is assuming that both $\lim_{n \rightarrow \infty} U(f, P_n)$ and $\lim_{n \rightarrow \infty} L(f, P_n)$ exist, which is not given and is part of what we need to prove.

We can prove it directly using the definition of sequential limit. We're given $\lim_{n \rightarrow \infty} [U(f, P_n) - L(f, P_n)] = 0$. Choose N large enough such that for every $n \geq N$

$$U(f, P_n) - L(f, P_n) < \epsilon$$

The same N works to prove that $\lim_{n \rightarrow \infty} U(f, P_n) = U(f)$ because $U(f, P) \geq U(f) \geq L(f, P)$ for any P , hence

$$U(f, P_n) - U(f) \leq U(f, P_n) - L(f, P_n) < \epsilon$$

A similar argument can be used to show that $\lim_{n \rightarrow \infty} L(f, P_n) = L(f)$. Now we can apply the Algebraic Limit Theorem to conclude that $\lim_{n \rightarrow \infty} [U(f, P_n) - L(f, P_n)] = 0 \implies \lim_{n \rightarrow \infty} U(f, P_n) = \lim_{n \rightarrow \infty} L(f, P_n)$.

(b)

$$U(f, P_n) = \frac{1}{n} \sum_{i=1}^n \frac{i}{n} = \frac{1}{n^2} \sum_{i=1}^n i = \frac{n+1}{2n}$$

$$L(f, P_n) = \frac{1}{n} \sum_{i=1}^n \frac{i-1}{n} = \frac{1}{n^2} \sum_{i=1}^n (i-1) = \frac{n+1}{2n} - \frac{1}{n}$$

(c) We have $\lim_{n \rightarrow \infty} U(f, P_n) - L(f, P_n) = \lim_{n \rightarrow \infty} \frac{1}{n} = 0$, so the sequential criterion is satisfied.

To compute $\int_0^1 f$, we note that

$$U(f, P_n) = \frac{n+1}{2n} = \frac{1}{2} + \frac{1}{2n}$$

Hence the area is

$$\lim_{n \rightarrow \infty} U(f, P_n) = \frac{1}{2}$$

Similarly, $\lim_{n \rightarrow \infty} L(f, P_n) = \frac{1}{2}$ as well, although we don't really need to check this since the sequential criterion guarantees that $\lim_{n \rightarrow \infty} U(f, P_n) = \lim_{n \rightarrow \infty} L(f, P_n)$.

Note that the result agrees with the geometric area as well; the area under $f(x) = x$ from $[0, 1]$ is a triangle with base and height 1, so its area is $\frac{1}{2}$.

4. Let g be bounded on $[a, b]$ and assume there exists a partition P with $L(g, P) = U(g, P)$. Describe g . Is g necessarily continuous? Is it integrable? If so, what is the value of $\int_a^b g$?

Solution: This can happen if and only if each segment of the partition is a constant i.e. for each segment $[x_{i-1}, x_i]$, we have $g(x) = c_i$. Otherwise one segment will have a supremum greater than infimum, causing $U(g, P) > L(g, P)$.

But note that two adjacent segments share an endpoint e.g. x_i is both in $[x_{i-1}, x_i]$ and $[x_i, x_{i+1}]$. This forces the whole interval to be a constant.

So g is just a constant function $g(x) = c$. It's continuous, integrable, and assuming the partition has length n , the integral is

$$\int_a^b g = \sum_{i=1}^n (x_i - x_{i-1})c = (b - a)c$$

5. Assume that for each n , f_n is an integrable function on $[a, b]$. If $(f_n) \rightarrow f$ uniformly on $[a, b]$, prove that f is also integrable on this set. (We will see that this conclusion does not necessarily follow if the convergence is pointwise).

Solution: We need to somehow relate the upper/lower sums of f_n with those of f . First, let's use the uniform convergence to choose an N such that for all $n \geq N$, $|f_n(x) - f(x)| < \epsilon$ for all $x \in [a, b]$.

Now take a partition $P = \{x_0, \dots, x_m\}$ and let's see how this affects $U(f_n, P)$ and $U(f, P)$. We have

$$U(f_n, P) = \sum_{i=1}^m (x_i - x_{i-1})M_{n,i}$$

where $M_{n,i}$ is the supremum of f_n in $[x_{i-1}, x_i]$

Since $f(x) < f_n(x) + \epsilon \leq M_{n,i} + \epsilon$, it follows that the supremum M_i of f is bounded by $M_i \leq M_{n,i} + \epsilon$. Then $U(f, P)$ is bounded by

$$U(f, P) \leq \sum_{i=1}^m (x_i - x_{i-1})(M_{n,i} + \epsilon) = U(f_n, P) + (b - a)\epsilon$$

We can adjust our initial choice of N so that $|f_n(x) - f(x)| < \frac{\epsilon}{4(b-a)}$ instead, which gives

$$U(f, P) \leq U(f_n, P) + \frac{\epsilon}{4}$$

Similarly, we can prove that $L(f, P) \geq L(f_n, P) - \frac{\epsilon}{4}$.

Our goal is to prove the integrability of f by Theorem 7.2.8: given an $\epsilon > 0$, produce a partition P_ϵ such that $U(f, P_\epsilon) - L(f, P_\epsilon) < \epsilon$.

Let's see what we have so far with an arbitrary partition P :

$$U(f, P) - L(f, P) \leq [U(f_n, P) + \frac{\epsilon}{4}] - [L(f_n, P) - \frac{\epsilon}{4}] = U(f_n, P) - L(f_n, P) + \frac{\epsilon}{2}$$

The remaining issue is to control $U(f_n, P) - L(f_n, P)$. Note that while this inequality holds for all $n \geq N$, to prove Theorem 7.2.8 we can pick a single specific f_n . By integrability of f_n , there exists a partition P_n such that

$$U(f_n, P_n) - L(f_n, P_n) < \frac{\epsilon}{2}$$

Set $P_\epsilon = P_n$, then

$$U(f, P_n) - L(f, P_n) \leq U(f_n, P_n) - L(f_n, P_n) + \frac{\epsilon}{2} < \frac{\epsilon}{2} + \frac{\epsilon}{2} = \epsilon$$

6. A *tagged partition* $(P, \{c_k\})$ is one where in addition to a partition P we choose a sampling point c_k in each of the subintervals $[x_{k-1}, x_k]$. The corresponding *Riemann sum* is defined as

$$R(f, P) = \sum_{k=1}^n f(c_k) \Delta x_k$$

Riemann's Original Definition of the Integral: A bounded function f is *integrable* on $[a, b]$ with $\int_a^b f = A$ if for all $\epsilon > 0$ there exists a $\delta > 0$ such that for any tagged partition $(P, \{c_k\})$ satisfying $\Delta x_k < \delta$ for all k , it follows that

$$|R(f, P) - A| < \epsilon$$

Show that if f satisfies Riemann's definition above, then f is integrable in the sense of Definition 7.2.7. (The full equivalence of these two characterizations of integrability is proved in Section 8.1).

Solution: My first thought was to choose c_k such that $f(c_k)$ is the supremum / infimum so that $R(f, P)$ becomes equal the upper/lower sum. But this doesn't work because there may be no point $c_k \in [x_{k-1}, x_k]$ such that $f(c_k)$ is the supremum.

Instead, we can choose c_k such that $f(c_k)$ is close to the supremum so that $U(f, P)$ is close to $R(f, P)$. Then control $R(f, P)$ so that it's close to A , then we can make $U(f, P)$ close to A . Do the same with $L(f, P)$, then we can make $U(f, P)$ close to $L(f, P)$ by controlling their distance to A .

More formally, we're going to prove this by Theorem 7.2.8: for any $\epsilon > 0$, produce a partition P_ϵ such that $U(f, P_\epsilon) - L(f, P_\epsilon) < \epsilon$.

Riemann's original definition works for *any* tagged partition $(P, \{c_k\})$ satisfying $\Delta x_k < \delta$. Pick a δ such that

$$|R(f, P) - A| < \frac{\epsilon}{4}$$

Now choose a specific partition $P_n = \{x_0, \dots, x_n\}$ with $\Delta x_k < \delta$ and with tags $\{c_k\}$ such that for all k

$$M_k - f(c_k) < \frac{\epsilon}{4(b-a)}$$

It's always possible to choose such $f(c_k)$ by virtue of M_k being a supremum. Then we have

$$\begin{aligned} U(f, P_n) - R(f, P_n) &= \sum_{k=1}^n (M_k - f(c_k)) \Delta x_k \\ &< \frac{\epsilon}{4(b-a)} \sum_{k=1}^n \Delta x_k = \frac{\epsilon}{4} \end{aligned}$$

Combining them, we get

$$|U(f, P_n) - A| \leq |U(f, P_n) - R(f, P_n)| + |R(f, P_n) - A| < \frac{\epsilon}{4} + \frac{\epsilon}{4} = \frac{\epsilon}{2}$$

Now for the lower sum. We can reuse the same partition P_n (it's important we do so for Theorem 7.2.8) but with a different sampling point d_k such that

$$f(d_k) - m_k < \frac{\epsilon}{4(b-a)}$$

And we do a similar computation as the upper sum to get

$$|L(f, P_n) - A| < \frac{\epsilon}{2}$$

And now we've satisfied Theorem 7.2.8. Let $P_\epsilon = P_n$, then

$$U(f, P_n) - L(f, P_n) \leq |U(f, P_n) - A| + |A - L(f, P_n)| < \frac{\epsilon}{2} + \frac{\epsilon}{2} = \epsilon$$

We can also prove that $U(f) = L(f) = A$. Theorem 7.2.8 already gives $U(f) = L(f)$. Let $U(f) = L(f) = I$, then

$$\begin{aligned} L(f, P_n) &\leq L(f) = I = U(f) \leq U(f, P_n) \\ \implies A - \frac{\epsilon}{2} &< I < A + \frac{\epsilon}{2} \\ \implies |I - A| &< \frac{\epsilon}{2} \end{aligned}$$

Since this works for any $\epsilon > 0$, by Theorem 1.2.6 $I = A$.

7. Let $f : [a, b] \rightarrow \mathbf{R}$ be increasing on the set $[a, b]$ (i.e. $f(x) \leq f(y)$ whenever $x \leq y$). Show that f is integrable on $[a, b]$.

Solution: Play around with how an increasing function affects $U(f, P)$ and $L(f, P)$ for a partition P . Some observations (in the order I noticed them):

1. As k increases, M_k and m_k increase as well so the upper / lower sum in the subinterval $[x_{k-1}, x_k]$ must also increase. But this observation doesn't seem to help.
2. The lower sum of $[x_k, x_{k+1}]$ must be equal or greater than the upper sum of $[x_{k-1}, x_k]$, as $m_{k+1} \geq M_k$. This doesn't seem to help either.
3. But (2) led me to realize that actually for any subinterval $[x_{k-1}, x_k]$, we have $m_k = f(x_{k-1})$ and $M_k = f(x_k)$. In particular, $m_k = M_{k-1}$.

Observation (3) strongly hints towards telescoping sum. We have

$$\begin{aligned} U(f, P) - L(f, P) &= (\Delta x_1 M_1 + \dots + \Delta x_n M_n) - (\Delta x_1 m_1 + \dots + \Delta x_n m_n) \\ &= [\Delta x_1 f(x_1) + \dots + \Delta x_n f(x_n)] - [\Delta x_1 f(x_0) + \dots + \Delta x_n f(x_{n-1})] \\ &= \sum_{k=1}^n \Delta x_k [f(x_k) - f(x_{k-1})] \end{aligned}$$

That didn't telescope as cleanly as I'd hoped due to the differing Δx_k coefficients, but it's still the right approach.

— **Incorrect attempt** —

Let $\Delta f_k = f(x_k) - f(x_{k-1})$. We can choose a partition P such that $\Delta x_k < \frac{\epsilon}{n \Delta f_k}$.

Then

$$\begin{aligned} U(f, P) - L(f, P) &= \sum_{k=1}^n \Delta x_k \Delta f_k \\ &< \sum_{k=1}^n \frac{\epsilon}{n \Delta f_k} \Delta f_k \\ &= \epsilon \end{aligned}$$

Satisfying the integrability criterion of Theorem 7.2.8.

(In the special case where $\Delta f_k = 0$, then $\Delta x_k \Delta f_k = 0$ so choose any number for the Δx_k).

— **Mistake pointed out by chatgpt. Correction my own.** —

The above doesn't work because the value of Δf_k itself depends on the chosen partition, so the argument is circular. The other issue is that $\sum_{k=1}^n \Delta x_k$ must sum up to $(b - a)$.

Instead, consider an equally-spaced partition (this is the same trick as exercise 2c). Let $L = (b - a)$ and $\Delta x_k = \frac{L}{n}$, then

$$U(f, P) - L(f, P) = \sum_{k=1}^n \frac{L}{n} \Delta f_k = \frac{L}{n} [f(b) - f(a)]$$

Then choose a partition with n large enough such that $\frac{L}{n} [f(b) - f(a)] < \epsilon$, satisfying the integrability criterion of Theorem 7.2.8.